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WO 2018/165048 (13.09.2018 Gazette 2018/37)(54) **SOLID FORMS AND COMBINATION COMPOSITIONS COMPRISING A BETA-LACTAMASE INHIBITOR AND USES THEREOF**

FESTE FORMEN UND KOMBINATIONSZUSAMMENSETZUNGEN MIT EINEM
 BETA-LACTAMASE-HEMMER UND VERWENDUNGEN DAVON

FORMES SOLIDES ET COMPOSITIONS DE COMBINAISON COMPRENANT UN INHIBITEUR DE
 BÊTA-LACTAMASE ET LEURS UTILISATIONS

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EP 3 592 362 B1

Description**FIELD OF INVENTION**

[0001] The present invention relates to pharmaceutical compositions containing boron-containing compounds, polymorphic forms, preparations and their use as inhibitors of beta-lactamase enzymes and as antibacterial agents in combination with a beta-lactam antibiotic.

BACKGROUND OF THE INVENTION

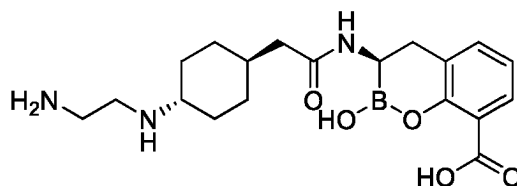
[0002] Antibiotics are the most effective drugs for treating bacteria-infectious diseases. They are largely used in the clinic because of their good antibacterial effect with limited side effects. Among them, the beta-lactam class of antibiotics (for example, penicillins, cephalosporins, monobactams and carbapenems) are preferred because their effect is bactericidal and their target is absent in eukaryotic cells with consequent low toxicity.

[0003] To counter the efficacy of the various beta-lactams, bacteria have evolved to produce variants of beta-lactam deactivating enzymes called beta-lactamases, and in the ability to share this tool both vertically and horizontally inter- and intra-species. These beta-lactamases are categorized as "serine" or "metallo" based, respectively, based on the presence of a key serine or zinc in the enzyme active site. The rapid induction, selection and spread of this mechanism of bacterial resistance can severely limit the whole class of beta-lactam treatment options in the hospital and in the community. There is a need for effective and safe therapeutic agents that can treat such resistant infections.

[0004] WO 2014/089365 A1 (Venatorx Pharmaceuticals Inc.; 12 June 2014) describes certain compounds and compositions that modulate the activity of beta-lactamases and their use in the treatment of bacterial infections.

SUMMARY OF THE INVENTION

[0005] One aspect of the invention is a crystalline form of (R)-3-(2-(trans-4-(2-aminoethylamino)cyclohexyl)acetamido)-2-hydroxy-3,4-dihydro-2H-benzo[e][1,2]oxaborinine-8-carboxylic acid dihydrochloride:



wherein the crystalline form is anhydrous; and

wherein the crystalline form has an X-ray powder diffraction (XRPD) pattern that includes at least three characteristic peaks selected from peaks at:

$7.3^{\circ} \pm 0.1^{\circ} 2\theta$, $10.8^{\circ} \pm 0.1^{\circ} 2\theta$, $14.5^{\circ} \pm 0.1^{\circ} 2\theta$, $16.6^{\circ} \pm 0.1^{\circ} 2\theta$, $18.0^{\circ} \pm 0.1^{\circ} 2\theta$, $19.6^{\circ} \pm 0.1^{\circ} 2\theta$, $19.7^{\circ} \pm 0.1^{\circ} 2\theta$, $23.3^{\circ} \pm 0.1^{\circ} 2\theta$, $24.0^{\circ} \pm 0.1^{\circ} 2\theta$, $24.3^{\circ} \pm 0.1^{\circ} 2\theta$, $27.3^{\circ} \pm 0.1^{\circ} 2\theta$, and $29.3^{\circ} \pm 0.1^{\circ} 2\theta$.

[0006] The crystalline form, (R)-3-(2-(trans-4-(2-aminoethylamino)cyclohexyl)acetamido)-2-hydroxy-3,4-dihydro-2H-benzo[e][1,2]oxaborinine-8-carboxylic acid dihydrochloride, is anhydrous.

[0007] In some embodiments, the crystalline form has an X-Ray powder diffraction (XRPD) pattern substantially the same as shown in FIG. 8.

[0008] In some embodiments, the crystalline form has an X-ray powder diffraction (XRPD) pattern comprising characteristic peaks at about $7.3^{\circ} \pm 0.1^{\circ} 2\theta$, about $14.5^{\circ} \pm 0.1^{\circ} 2\theta$, about $18.0^{\circ} \pm 0.1^{\circ} 2\theta$, about $19.7^{\circ} \pm 0.1^{\circ} 2\theta$, about $24.0^{\circ} \pm 0.1^{\circ} 2\theta$, and about $27.3^{\circ} \pm 0.1^{\circ} 2\theta$.

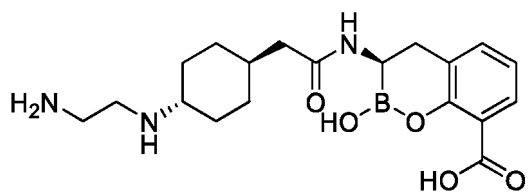
[0009] In some embodiments of the crystalline form, the X-ray powder diffraction (XRPD) pattern further comprises characteristic peaks at about $10.8^{\circ} \pm 0.1^{\circ} 2\theta$, about $16.6^{\circ} \pm 0.1^{\circ} 2\theta$, about $19.6^{\circ} \pm 0.1^{\circ} 2\theta$, about $23.3^{\circ} \pm 0.1^{\circ} 2\theta$, about $24.3^{\circ} \pm 0.1^{\circ} 2\theta$, and about $29.3^{\circ} \pm 0.1^{\circ} 2\theta$.

[0010] Another aspect of the invention is a pharmaceutical composition comprising:

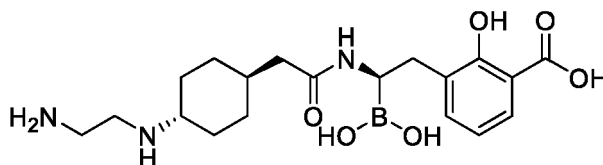
- (i) a crystalline form of the first aspect; and
- (ii) cefepime.

[0011] In some embodiments of a pharmaceutical composition, the pharmaceutical composition is formulated as a homogeneous liquid suitable for injection.

zo[e][1,2]oxaborinine-8-carboxylic acid is as shown in the structure below:



(R)-3-(2-(trans-4-(2-aminoethylamino)cyclohexyl)acetamido)-2-hydroxy-3,4-dihydro-2H-benzo[e][1,2]oxaborinine-8-carboxylic acid is also referred to as Compound 1. Compound 1 may exist in an equilibrium between the "closed" cyclic form (as shown above) and the "open" acyclic form:



Compound 1 may associate into intramolecular dimers, trimers, and any combinations thereof.

Crystalline Forms

[0025] Described herein are crystalline forms of (R)-3-(2-(trans-4-(2-aminoethylamino)cyclohexyl)acetamido)-2-hydroxy-3,4-dihydro-2H-benzo[e][1,2]oxaborinine-8-carboxylic acid dihydrochloride.

Reference Form 1

[0026] The dihydrochloride salt of Compound 1 can be made into a pure and stable crystalline form (Form 1) under controlled conditions. In some embodiments, Form 1 is a monohydrate form of the dihydrochloride salt of Compound 1. In some embodiments, the X-ray powder diffraction (XRPD) pattern of Form 1 is substantially the same as shown in FIG. 2, with corresponding tabulated peak data shown in Table 1. In some embodiments, the X-ray powder diffraction (XRPD) pattern of Form 1 is substantially the same as shown in FIG. 3. The X-ray powder diffraction (XRPD) pattern shown in FIG. 2 is substantially the same as the X-ray powder diffraction (XRPD) pattern shown in FIG. 3.

Table 1

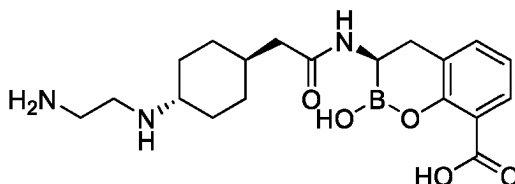
2 θ [°]	d value [Å]	Intensity [%]	2 θ [°]	d value [Å]	Intensity [%]
7.0	12.54	19	28.6	3.12	25
8.6	10.29	25	28.8	3.09	13
10.5	8.39	88	29.6	3.02	21
11.2	7.93	12	30.1	2.97	14
12.7	6.96	42	30.3	2.94	17
13.1	6.73	12	31.2	2.87	12
13.6	6.50	17	31.4	2.85	14
14.1	6.28	16	31.5	2.84	13
14.9	5.93	30	31.9	2.80	18
16.1	5.51	68	32.7	2.73	16
16.9	5.24	18	32.8	2.73	15
17.2	5.15	11	33.0	2.71	27
17.6	5.02	14	33.4	2.68	15
18.9	4.69	52	33.6	2.66	12

Table 6

2 θ [°]	d value [Å]	Intensity [%]
8.74	10.11	100
8.81	10.02	72.2
13.38	6.61	48.8
16.30	5.43	49.1
20.06	4.42	77.6
22.86	3.89	87.7
28.30	3.15	44.7

Embodiments

[0033] A first aspect of the invention is a crystalline form of (R)-3-(2-(trans-4-(2-aminoethylamino)cyclohexyl)acetamido)-2-hydroxy-3,4-dihydro-2H-benzo[e][1,2]oxaborinine-8-carboxylic acid dihydrochloride:



wherein the crystalline form is anhydrous; and

wherein the crystalline form has an X-ray powder diffraction (XRPD) pattern that includes at least three characteristic peaks selected from peaks at:

$7.3^{\circ} \pm 0.1^{\circ} 2\theta$, $10.8^{\circ} \pm 0.1^{\circ} 2\theta$, $14.5^{\circ} \pm 0.1^{\circ} 2\theta$, $16.6^{\circ} \pm 0.1^{\circ} 2\theta$, $18.0^{\circ} \pm 0.1^{\circ} 2\theta$, $19.6^{\circ} \pm 0.1^{\circ} 2\theta$, $19.7^{\circ} \pm 0.1^{\circ} 2\theta$, $23.3^{\circ} \pm 0.1^{\circ} 2\theta$, $24.0^{\circ} \pm 0.1^{\circ} 2\theta$, $24.3^{\circ} \pm 0.1^{\circ} 2\theta$, $27.3^{\circ} \pm 0.1^{\circ} 2\theta$, and $29.3^{\circ} \pm 0.1^{\circ} 2\theta$.

[0034] The crystalline form of (R)-3-(2-(trans-4-(2-aminoethylamino)cyclohexyl)acetamido)-2-hydroxy-3,4-dihydro-2H-benzo[e][1,2]oxaborinine-8-carboxylic acid dihydrochloride is anhydrous.

[0035] In some embodiments, the crystalline form has at least one of the following properties:

(a) an X-Ray powder diffraction (XRPD) pattern substantially the same as shown in FIG. 8; or

(b) an X-ray powder diffraction (XRPD) pattern comprising characteristic peaks at about $7.3^{\circ} \pm 0.1^{\circ} 2\theta$, about $14.5^{\circ} \pm 0.1^{\circ} 2\theta$, about $18.0^{\circ} \pm 0.1^{\circ} 2\theta$, about $19.7^{\circ} \pm 0.1^{\circ} 2\theta$, about $24.0^{\circ} \pm 0.1^{\circ} 2\theta$, and about $27.3^{\circ} \pm 0.1^{\circ} 2\theta$.

[0036] In some embodiments, the X-ray powder diffraction (XRPD) pattern further comprises characteristic peaks at about $10.8^{\circ} \pm 0.1^{\circ} 2\theta$, about $16.6^{\circ} \pm 0.1^{\circ} 2\theta$, about $19.6^{\circ} \pm 0.1^{\circ} 2\theta$, about $23.3^{\circ} \pm 0.1^{\circ} 2\theta$, about $24.3^{\circ} \pm 0.1^{\circ} 2\theta$, and about $29.3^{\circ} \pm 0.1^{\circ} 2\theta$.

[0037] In some embodiments, the crystalline form is Form 2.

Pharmaceutical Compositions

[0038] Again, any references herein to methods of treatment refer to the compounds, pharmaceutical compositions, and medicaments of the present invention for use in a such methods.

[0039] A second aspect of the invention is a pharmaceutical composition comprising:

- (i) a crystalline form of the first aspect; and
- (ii) cefepime.

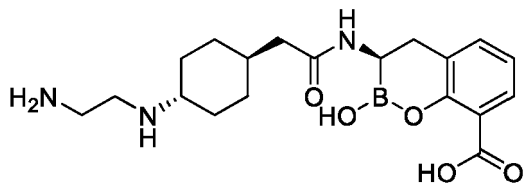
In some embodiments, the crystalline form of (R)-3-(2-(trans-4-(2-aminoethylamino)cyclohexyl)acetamido)-2-hydroxy-3,4-dihydro-2H-benzo[e][1,2]oxaborinine-8-carboxylic acid dihydrochloride is Form 2.

[0040] In some embodiments, the pharmaceutical compositions described herein are provided in unit dosage form. As used herein, a "unit dosage form" is a composition containing an amount of compound 1 that is suitable for administration to

Embodiments

[0052] A second aspect of the invention is a pharmaceutical composition comprising:

- (i) (R)-3-(2-(trans-4-(2-aminoethylamino)cyclohexyl)acetamido)-2-hydroxy-3,4-dihydro-2H-benzo[e][1,2]oxaborinine-8-carboxylic acid dihydrochloride:



wherein the crystalline form is anhydrous; and
 wherein the crystalline form has an X-ray powder diffraction (XRPD) pattern that includes at least three characteristic peaks selected from peaks at: $7.3^{\circ} \pm 0.1^{\circ}$ 2 θ , $10.8^{\circ} \pm 0.1^{\circ}$ 2 θ , $14.5^{\circ} \pm 0.1^{\circ}$ 2 θ , $16.6^{\circ} \pm 0.1^{\circ}$ 2 θ , $18.0^{\circ} \pm 0.1^{\circ}$ 2 θ , $19.6^{\circ} \pm 0.1^{\circ}$ 2 θ , $19.7^{\circ} \pm 0.1^{\circ}$ 2 θ , $23.3^{\circ} \pm 0.1^{\circ}$ 2 θ , $24.0^{\circ} \pm 0.1^{\circ}$ 2 θ , $24.3^{\circ} \pm 0.1^{\circ}$ 2 θ ; and

- (ii) cefepime.

[0053] In some embodiments, the pharmaceutical composition is formulated as a homogeneous liquid suitable for injection.

[0054] In some embodiments, the pharmaceutical composition further comprises a pharmaceutically acceptable excipient.

[0055] In some embodiments, the pharmaceutically acceptable excipient is selected from amino acids.

[0056] In some embodiments, the amino acid is L-arginine.

[0057] In some embodiments, the pharmaceutical composition further comprises an aqueous carrier.

[0058] In some embodiments, the aqueous carrier is selected from water for injection, 0.9% sodium chloride injection, 5% dextrose injection, 10% dextrose injection, sodium lactate injection, 5% dextrose and 0.9% sodium chloride injection, lactated Ringers and 5% dextrose injection, sodium chloride/sodium acetate/sodium gluconate/potassium chloride/magnesium chloride injection, sodium chloride/potassium acetate/magnesium acetate in 5% dextrose injection, and any combinations thereof.

[0059] In some embodiments, the pharmaceutical composition has a pH from about 4 to about 9.

[0060] In some embodiments, the pharmaceutical composition has a pH from about 4 to about 6.

[0061] In some embodiments the pharmaceutical composition is formulated as a powder for reconstitution.

[0062] In some embodiments, the pharmaceutical composition further comprises a pharmaceutically acceptable excipient.

[0063] In some embodiments, the pharmaceutically acceptable excipient is selected from amino acids.

[0064] In some embodiments, the amino acid is L-arginine.

[0065] In some embodiments, the pharmaceutical composition is suitable for injection once reconstituted with an aqueous carrier.

[0066] In some embodiments, the aqueous carrier is selected from water for injection, 0.9% sodium chloride injection, 5% dextrose injection, 10% dextrose injection, sodium lactate injection, 5% dextrose and 0.9% sodium chloride injection, lactated Ringers and 5% dextrose injection, sodium chloride/sodium acetate/sodium gluconate/potassium chloride/magnesium chloride injection, sodium chloride/potassium acetate/magnesium acetate in 5% dextrose injection, and any combinations thereof.

[0067] In some embodiments, the pharmaceutical composition is stable at about $5^{\circ}\text{C} \pm 3^{\circ}\text{C}$ for at least 12 months.

[0068] In some embodiments, the pharmaceutical composition is stable at about $25^{\circ}\text{C} \pm 2^{\circ}\text{C}/60\% \text{ RH} \pm 5\% \text{ RH}$ for at least 12 months.

[0069] In some embodiments, the pharmaceutical composition is stable at about $30^{\circ}\text{C} \pm 2^{\circ}\text{C}/65\% \text{ RH} \pm 5\% \text{ RH}$ for at least 6 months.

[0070] In some embodiments, the pharmaceutical composition is stable at about $40^{\circ}\text{C} \pm 2^{\circ}\text{C}/75\% \text{ RH} \pm 5\% \text{ RH}$ for at least 6 months.

[0071] Also described herein is a method of administering to a subject in need thereof, a pharmaceutical composition comprising:

(Mettler-Toledo GmbH, Switzerland). The DSC822e was calibrated for temperature and enthalpy with a small piece of indium (melting point at 156.6 °C; $\Delta H_f = 28.45 \text{ J.g}^{-1}$). Samples were sealed in standard 40 μl aluminum pans, pin-holed and heated in the DSC from 25 °C to 300 °C, at a heating rate of 10 °C min⁻¹. Dry N₂ gas, at a flow rate of 50 ml min⁻¹ was used to purge the DSC equipment during measurement.

Thermogravimetric analysis coupled with mass spectroscopy (TGMS)

[0144] Mass loss due to solvent or water loss from the crystals was determined by TGA/SDTA (TGA: thermogravimetric analysis; SDTA: single differential thermal analysis). Monitoring the sample weight, during heating in a TGA/SDTA851e instrument (Mettler-Toledo GmbH, Switzerland), resulted in a weight vs. temperature curve. The TGA/SDTA851e was calibrated for temperature with indium and aluminum. Samples were weighed into 100 μl aluminum crucibles and sealed. The seals were pin-holed and the crucibles heated in the TGA from 25 to 300 or 400 °C (depending on the experiment) at a heating rate of 10 °C min⁻¹. Dry N₂ gas was used for purging.

[0145] The gases evolved from the TGA samples were analyzed by a mass spectrometer Omnistar GSD 301 T2 (Pfeiffer Vacuum GmbH, Germany). The latter is a quadrupole mass spectrometer, which analyzes masses in the range of 0-200 amu.

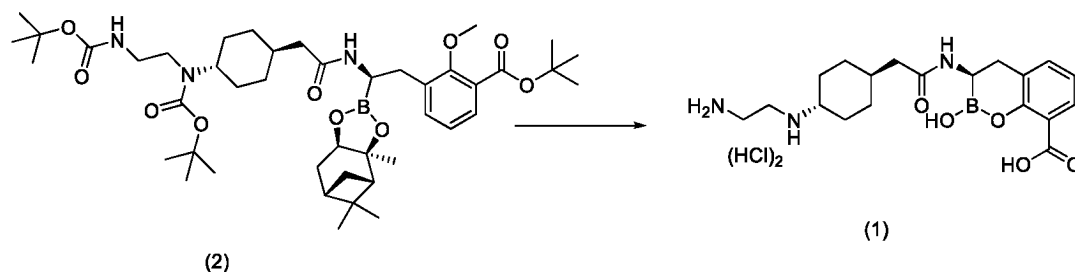
Dynamic vapor sorption (DVS) analysis

[0146] Differences in hygroscopicity (moisture uptake) of the various forms of a solid material provided a measure of their relative stability at increasing relative humidity. Moisture sorption isothermal profiles of small samples were obtained using a DVS-1 system from Surface Measurement Systems (London, UK); this instrument is suitable for use with as little as a few milligrams of sample, with an accuracy of 0.1 μg .

[0147] The following humidity profile was applied: The relative humidity was cycled from 45% to 95% (sorption), back to 0% (desorption) and to 45%, at a constant temperature of 25 °C. Steps consisted of 10% RH (between 45% and 75% RH), 5% RH (between 0 and 45% RH) and 20% (between 75% and 95% RH). Weight equilibration per step was set with a minimum holding time of 1 hour. At the end of the DVS experiments, the recovered solid material was measured by HT-XRPD.

Example 1: Preparation of Compound 1-dihydrochloride Salt

[0148]



[0149] To a solution of *tert*-butyl 3-((*R*)-2-(2-((1*r*,4*R*)-4-((*tert*-butoxycarbonyl)(2-((*tert*-butoxycarbonyl)amino)ethyl)amino)cyclohexyl)acetamido)-2-((3*aS*,4*S*,6*S*,7*aR*)-3*a*,5,5-trimethylhexahydro-4,6-methanobenzo[d][1,3,2]dioxaborol-2-yl)ethyl)-2-methoxybenzoate (Structure 2, 110.9 g, 136.7 mmol) in 1,4 dioxane (137 mL) was added hydrochloric acid (540 mL, 3M in water). The resulting mixture was heated to reflux and stirred at this temperature for 100 minutes. The resulting mixture was cooled to room temperature over 30 minutes then extracted with methyl *tert*-butyl ether (3x200 mL). The aqueous phase was concentrated under vacuum (57-29 mmHg and 20-72 °C external bath temperature, wherein 1 mmHg = 133.322 Pa). The residue was diluted to a total weight of 130g with water. To this solution was added isopropanol (90 mL) over approximately 10 minutes. This solution was seeded with Compound 1-(HCl)₂ crystals (100 mg). To this mixture was added isopropanol (100 mL) over approximately 5 minutes. The resulting slurry was stirred for approximately 15 hours then filtered under vacuum, the solid was washed with isopropanol (2x200 mL) then methyl *tert*-butyl ether (400 mL). The solid was further dried under high vacuum to give the title compound (structure 1) (43.9 g) as a white crystalline solid as a mixture of Form 1 and Form 2.

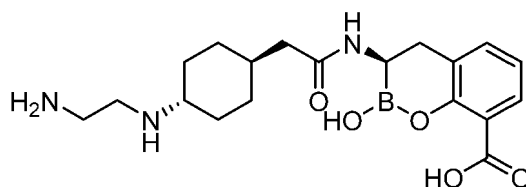
Table 22

Dosage Regimens	Subject Population	Treatment Duration
Compound 1 500 mg in combination with cefepime 2 g IV q8h	patients with cUTI	5-14 days
Compound 1 500 mg in combination with cefepime 2 g IV q8h	patients with cIAI	5-14 days
Compound 1 500 mg in combination with cefepime 2 g IV q8h	patients infected with resistant pathogens at multiple sites	5-14 days
Compound 1 750 mg in combination with cefepime 2 g IV q8h	patients with cUTI	5-14 days
Compound 1 750 mg in combination with cefepime 2 g IV q8h	patients with cIAI	5-14 days
Compound 1 750 mg in combination with cefepime 2 g IV q8h	patients infected with resistant pathogens at multiple sites	5-14 days
Abbreviations: ~≈approximately; cIAI=complicated intra-abdominal infection; cUTI=complicated urinary tract infection; hr=hour; IV=intravenous; q8h=three times daily, 8 hours apart;		

[0220] Treatments in Table 22 are administered every 8 hours (q8h) as 2-hour IV infusion for 10 days. Subjects will receive a total of 28 doses with a single dose administered on Day 10.

Claims

1. A crystalline form of (R)-3-(2-(trans-4-(2-aminoethylamino)cyclohexyl)acetamido)-2-hydroxy-3,4-dihydro-2H-benzo[e][1,2]oxaborinine-8-carboxylic acid dihydrochloride:



wherein the crystalline form is anhydrous; and

wherein the crystalline form has an X-ray powder diffraction (XRPD) pattern that includes at least three characteristic peaks selected from peaks at:

$7.3^{\circ} \pm 0.1^{\circ} 2\theta$, $10.8^{\circ} \pm 0.1^{\circ} 2\theta$, $14.5^{\circ} \pm 0.1^{\circ} 2\theta$, $16.6^{\circ} \pm 0.1^{\circ} 2\theta$, $18.0^{\circ} \pm 0.1^{\circ} 2\theta$, $19.6^{\circ} \pm 0.1^{\circ} 2\theta$, $19.7^{\circ} \pm 0.1^{\circ} 2\theta$, $23.3^{\circ} \pm 0.1^{\circ} 2\theta$, $24.0^{\circ} \pm 0.1^{\circ} 2\theta$, $24.3^{\circ} \pm 0.1^{\circ} 2\theta$, $27.3^{\circ} \pm 0.1^{\circ} 2\theta$, and $29.3^{\circ} \pm 0.1^{\circ} 2\theta$.

2. The crystalline form of claim 1, wherein the crystalline form has an X-ray powder diffraction (XRPD) pattern comprising characteristic peaks at:

$7.3^{\circ} \pm 0.1^{\circ} 2\theta$, $14.5^{\circ} \pm 0.1^{\circ} 2\theta$, $18.0^{\circ} \pm 0.1^{\circ} 2\theta$, $19.7^{\circ} \pm 0.1^{\circ} 2\theta$, $24.0^{\circ} \pm 0.1^{\circ} 2\theta$, and $27.3^{\circ} \pm 0.1^{\circ} 2\theta$.

3. The crystalline form of claim 2, wherein the X-ray powder diffraction (XRPD) pattern further comprises characteristic peaks at:

$10.8^{\circ} \pm 0.1^{\circ} 2\theta$, $16.6^{\circ} \pm 0.1^{\circ} 2\theta$, $19.6^{\circ} \pm 0.1^{\circ} 2\theta$, $23.3^{\circ} \pm 0.1^{\circ} 2\theta$, $24.3^{\circ} \pm 0.1^{\circ} 2\theta$, and $29.3^{\circ} \pm 0.1^{\circ} 2\theta$.

4. A pharmaceutical composition comprising:

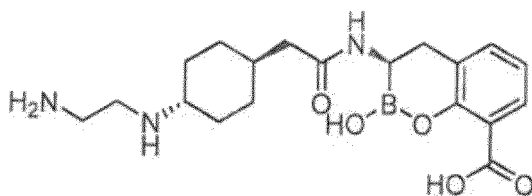
- (i) a crystalline form of any one of claims 1-3; and
- (ii) cefepime.

5. The pharmaceutical composition of claim 4, wherein the pharmaceutical composition is formulated as a powder for reconstitution.

6. The pharmaceutical composition of claim 4 or 5, further comprising a pharmaceutically acceptable excipient.
7. The pharmaceutical composition of claim 6, wherein the pharmaceutically acceptable excipient is L-arginine.
8. The pharmaceutical composition of any one of claims 4-7, wherein the pharmaceutical composition comprises 100 to 2000 mg of (R)-3-(2-(trans-4-(2-aminoethylamino)cyclohexyl)acetamido)-2-hydroxy-3,4-dihydro-2H-benzo[e][1,2]oxaborinine-8-carboxylic acid.
9. The pharmaceutical composition of any one of claims 4-8, wherein the pharmaceutical composition comprises 250 to 2000 mg of cefepime.
10. The pharmaceutical composition of any one of claims 4-8, wherein the pharmaceutical composition comprises 2000 mg of cefepime.
11. The pharmaceutical composition of any one of claims 4-7, wherein the pharmaceutical composition comprises 500 mg of (R)-3-(2-(trans-4-(2-aminoethylamino)cyclohexyl)acetamido)-2-hydroxy-3,4-dihydro-2H-benzole[[1,2]oxaborinine-8-carboxylic acid and 2 g of cefepime.
12. The pharmaceutical composition of any one of claims 4-7, wherein the pharmaceutical composition comprises 750 mg of (R)-3-(2-(trans-4-(2-aminoethylamino)cyclohexyl)acetamido)-2-hydroxy-3,4-dihydro-2H-benzole[[1,2]oxaborinine-8-carboxylic acid and 2 g of cefepime.
13. A pharmaceutical composition of any one of claims 4-12 for use in a method of treating a bacterial infection in a subject.

Patentansprüche

1. Kristalline Form von (R)-3-(2-(trans-4-(2-Aminoethylamino)cyclohexyl)acetamido)-2-hydroxy-3,4-dihydro-2H-benzo[e][1,2]oxaborinin-8-carbonsäure-dihydrochlorid



wobei die kristalline Form wasserfrei ist; und

wobei die kristalline Form ein Röntgenbeugungs- (XRPD-) Muster aufweist, das zumindest drei charakteristische Reflexe umfasst, die aus folgenden Reflexen ausgewählt sind:

$7,3^\circ \pm 0,1^\circ 2\theta$, $10,8^\circ \pm 0,1^\circ 2\theta$, $14,5^\circ \pm 0,1^\circ 2\theta$, $16,6^\circ \pm 0,1^\circ 2\theta$, $18,0^\circ \pm 0,1^\circ 2\theta$, $19,6^\circ \pm 0,1^\circ 2\theta$, $19,7^\circ \pm 0,1^\circ 2\theta$, $23,3^\circ \pm 0,1^\circ 2\theta$, $24,0^\circ \pm 0,1^\circ 2\theta$, $24,3^\circ \pm 0,1^\circ 2\theta$, $27,3^\circ \pm 0,1^\circ 2\theta$ und $29,3^\circ \pm 0,1^\circ 2\theta$.

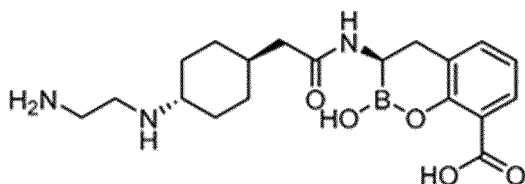
2. Kristalline Form nach Anspruch 1, wobei die kristalline Form ein Röntgenpulverbeugungs- (XRPD-) Muster, das charakteristische Peaks bei den folgenden Winkeln umfasst: $7,3^\circ \pm 0,1^\circ 2\theta$, $14,5^\circ \pm 0,1^\circ 2\theta$, $18,0^\circ \pm 0,1^\circ 2\theta$, $19,7^\circ \pm 0,1^\circ 2\theta$, $24,0^\circ \pm 0,1^\circ 2\theta$ und $27,3^\circ \pm 0,1^\circ 2\theta$.
3. Kristalline Form nach Anspruch 2, wobei das Röntgenpulverbeugungs- (XRPD-) Muster weitere charakteristische Peaks bei den folgenden Winkeln umfasst: $10,8^\circ \pm 0,1^\circ 2\theta$, $16,6^\circ \pm 0,1^\circ 2\theta$, $19,6^\circ \pm 0,1^\circ 2\theta$, $23,3^\circ \pm 0,1^\circ 2\theta$, $24,3^\circ \pm 0,1^\circ 2\theta$ und $29,3^\circ \pm 0,1^\circ 2\theta$.
4. Pharmazeutische Zusammensetzung, die Folgendes umfasst:
 - (i) eine kristalline Form nach einem der Ansprüche 1 bis 3; und
 - (ii) Cefepim.
5. Pharmazeutische Zusammensetzung nach Anspruch 4, wobei die pharmazeutische Zusammensetzung als Pulver

zur Rekonstitution formuliert ist.

6. Pharmazeutische Zusammensetzung nach Anspruch 4 oder 5, die weiters einen pharmazeutisch annehmbaren Hilfsstoff umfasst.
7. Pharmazeutische Zusammensetzung nach Anspruch 6, wobei der pharmazeutisch annehmbare Hilfsstoff L-Arginin ist.
8. Pharmazeutische Zusammensetzung nach einem der Ansprüche 4 bis 7, wobei die pharmazeutische Zusammensetzung 100 bis 2.000 mg (R)-3-(2-(trans-4-(2-Aminoethyl-amino)cyclohexyl)acetamido)-2-hydroxy-3,4-dihydro-2H-benzo[e][1,2]oxaborinin-8-carbonsäure umfasst.
9. Pharmazeutische Zusammensetzung nach einem der Ansprüche 4 bis 8, wobei die pharmazeutische Zusammensetzung 250 bis 2.000 mg Cefepim umfasst.
10. Pharmazeutische Zusammensetzung nach einem der Ansprüche 4 bis 8, wobei die pharmazeutische Zusammensetzung 2.000 mg Cefepim umfasst.
11. Pharmazeutische Zusammensetzung nach einem der Ansprüche 4 bis 7, wobei die pharmazeutische Zusammensetzung 500 mg (R)-3-(2-(trans-4-(2-Aminoethylamino)-cyclohexyl)acetamido)-2-hydroxy-3,4-dihydro-2H-benzo[e][1,2]oxaborinin-8-carbonsäure und 2 g Cefepim umfasst.
12. Pharmazeutische Zusammensetzung nach einem der Ansprüche 4 bis 7, wobei die pharmazeutische Zusammensetzung 750 mg (R)-3-(2-(trans-4-(2-Aminoethylamino)-cyclohexyl)acetamido)-2-hydroxy-3,4-dihydro-2H-benzo[e][1,2]oxaborinin-8-carbonsäure und 2 g Cefepim umfasst.
13. Pharmazeutische Zusammensetzung nach einem der Ansprüche 4 bis 12 zur Verwendung in einem Verfahren zur Behandlung einer bakteriellen Infektion bei einem Individuum.

Revendications

1. Forme cristalline de dihydrochlorure d'acide (R)-3-(2-(trans-4-(2-aminoéthylamino)cyclohexyl)acétamido)-2)-2-hydroxy-3,4-dihydro-2H-benzo[e][1,2]oxaborinine-8-carboxylique :



dans laquelle la forme cristalline est anhydre ; et

dans laquelle la forme cristalline présente un diagramme de diffraction de rayons X sur poudre (XRPD) qui inclut au moins trois pics caractéristiques sélectionnés parmi les pics à : $7,3^{\circ} \pm 0,1^{\circ} 2\theta$, $10,8^{\circ} \pm 0,1^{\circ} 2\theta$, $14,5^{\circ} \pm 0,1^{\circ} 2\theta$, $16,6^{\circ} \pm 0,1^{\circ} 2\theta$, $18,0^{\circ} \pm 0,1^{\circ} 2\theta$, $19,6^{\circ} \pm 0,1^{\circ} 2\theta$, $19,7^{\circ} \pm 0,1^{\circ} 2\theta$, $23,3^{\circ} \pm 0,1^{\circ} 2\theta$, $24,0^{\circ} \pm 0,1^{\circ} 2\theta$, $24,3^{\circ} \pm 0,1^{\circ} 2\theta$, $27,3^{\circ} \pm 0,1^{\circ} 2\theta$ et $29,3^{\circ} \pm 0,1^{\circ} 2\theta$.

2. Forme cristalline selon la revendication 1, dans laquelle la forme cristalline présente un diagramme de diffraction de rayons X sur poudre (XRPD) comprenant des pics caractéristiques à : $7,3^{\circ} \pm 0,1^{\circ} 2\theta$, $14,5^{\circ} \pm 0,1^{\circ} 2\theta$, $18,0^{\circ} \pm 0,1^{\circ} 2\theta$, $19,7^{\circ} \pm 0,1^{\circ} 2\theta$, $24,0^{\circ} \pm 0,1^{\circ} 2\theta$ et $27,3^{\circ} \pm 0,1^{\circ} 2\theta$.
3. Forme cristalline selon la revendication 2, dans laquelle le diagramme de diffraction de rayons X sur poudre (XRPD) comprend en outre des pics caractéristiques à : $10,8^{\circ} \pm 0,1^{\circ} 2\theta$, $16,6^{\circ} \pm 0,1^{\circ} 2\theta$, $19,6^{\circ} \pm 0,1^{\circ} 2\theta$, $23,3^{\circ} \pm 0,1^{\circ} 2\theta$, $24,3^{\circ} \pm 0,1^{\circ} 2\theta$ et $29,3^{\circ} \pm 0,1^{\circ} 2\theta$.
4. Composition pharmaceutique, comprenant :